BHOGYAM TEJASWINI

ASSIGNMENT – 01

QUESTIONS:

Section A: Control Structures

1. **Write a program to check whether a number is even or odd using an if-else statement.**
2. **Write a program to find the largest of three numbers using if-elif-else.**
3. **Write a program to print numbers from 1 to 50 using a for loop.**
4. **Write a program to calculate the factorial of a number using a while loop.**
5. **Write a program to display the multiplication table of a given number.**
6. **Write a program to check whether a number is prime or not.**
7. **Write a program to print the sum of digits of a number.**
8. **Write a program to reverse a number using a loop.**
9. **Write a program to print all even numbers between 1 and 100.**
10. **Write a program demonstrating the use of break and continue statements.**

Section B: List

1. **Write a program to find the largest element in a list**
2. **Write a program to remove duplicate elements from a list.**
3. **Write a program to sort a list in ascending and descending order.**
4. **Write a program to count the number of occurrences of an element in a list.**
5. **Write a program to merge two lists into one.**

Section C: Tuple

1. **Write a program to create a tuple and access its elements.**
2. **Write a program to find the length of a tuple.**
3. **Write a program to convert a tuple into a list.**
4. **Write a program to find the maximum and minimum values in a tuple.**
5. **Write a program to demonstrate tuple unpacking.**

Section D: Set

1. **Write a program to create a set and add elements to it.**
2. **Write a program to perform union, intersection, and difference between two sets.**
3. **Write a program to remove duplicates from a list using a set.**
4. **Write a program to check whether an element exists in a set.**
5. **Write a program to find common elements between two lists using sets.**

Answers: (Section A)

1.
```
num = int(input("Enter a number: "))
 if num % 2 == 0:
print(f"{num} is Even")
else:
print(f"{num} is Odd")
```

2.
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
print("Largest number is:", a)
elif b >= a and b >= c:
print("Largest number is:", b)
else:
print("Largest number is:", c)
```

3.
```
for i in range(1, 51):
  print(i)
```

4.
```
num = int(input("Enter a number: "))
fact = 1
i = 1
while i <= num:
   fact = fact * i
   i += 1
print("Factorial is:", fact)
```

5.
```
num = int(input("Enter a number: "))
for i in range(1, 11):
print(num, "x", i, "=", num * i)
```

6.
```
num = int(input("Enter a number: "))
is_prime = True
if num <= 1:
   is_prime = False
```

```
else:
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
if is_prime:
    print("Prime number")
else:
    print("Not a prime number")
```

7.
```
num = int(input("Enter a number: "))
sum = 0
while num > 0:
    digit = num % 10
    sum += digit
    num = num // 10
print("Sum of digits:", sum)
```

8.
```
num = int(input("Enter a number: "))
rev = 0
while num > 0:
    digit = num % 10
    rev = rev * 10 + digit
    num = num // 10
print("Reversed number:", rev)
```

9.
```
for i in range(1, 101):
if i % 2 == 0:
print(i)
```

10.
```
# break example
for i in range(1, 10):
    if i == 5:
        break
    print(i)
print("----")
# continue example
for i in range(1, 10):
    if i == 5:
        continue
    print(i)
```

(Section B):

1.
```
numbers = [10, 25, 5, 40, 15]
largest = numbers[0]
for num in numbers:
if num > largest:
largest = num
print("Largest element:", largest)
```

2.
```
numbers = [1, 2, 2, 3, 4, 4, 5]
unique = []
for num in numbers:
if num not in unique:
unique.append(num)
print("List without duplicates:", unique)
```

3.
```
numbers = [5, 2, 9, 1, 7]
numbers.sort()
print("Ascending:", numbers)
numbers.sort(reverse=True)
print("Descending:", numbers)
```

4.
```
numbers = [1, 2, 3, 2, 4, 2, 5]
element = int(input("Enter element to count: "))
count = 0
for num in numbers:
if num == element:
 count += 1
print("Occurrences:", count)
```

5.
```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
merged = list1 + list2
print("Merged list:", merged)
```

(Section C):

1.
```
t = (10, 20, 30, 40)
print("Tuple:", t)
print("First element:", t[0])
print("Second element:", t[1])
```

2.
```
t = (1, 2, 3, 4, 5)
print("Length of tuple:", len(t))
```

3.
```
t = (10, 20, 30)
l = list(t)
print("Tuple:", t)
print("List:", l)
```

4.
```
t = (5, 2, 9, 1, 7)
print("Maximum:", max(t))
print("Minimum:", min(t))
```

5.
```
t = (50, 90, 20)
a, b, c = t
print("a =", a)
print("b =", b)
print("c =", c)
```

Section D:

1.
```
s = {23, 42, 53}
s.add(41)
s.add(57)
print(“set:”,s)
```

2.
```
set1 = {11, 12, 13}
set2 = {13, 14, 15}
print("Union:", set1 | set2)
print("Intersection:", set1 & set2)
print("Difference:", set1 - set2)
```

3.
```
numbers = [21, 2, 2, 33, 54]
unique = list(set(numbers))
print("Without duplicates:", unique)
```

4.
```
s = {10, 20, 30, 40}
num = int(input("Enter element: "))
if num in s:
    print("Element exists")
else:
    print("Element not found")
```

5.
```
list1 = [31, 22, 53, 24]
list2 = [31, 4, 5, 6]
common = set(list1) & set(list2)
print("Common elements:", list(common))
```

(Section E):

1.
```
student = {"name": "Tejaswini", "age": 22, "course": "CSE"}
print("Name:", student["name"])
print("Age:", student["age"])
```

2.
```
text = input("Enter a string: ")
freq = {}
for ch in text:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1
print("Character frequency:", freq)
```

3.
```
d1 = {"a": 22, "b": 26}
d2 = {"c": 34, "d": 42}
d1.update(d2)
print("Merged dictionary:", d1)
```

4.
```
d = {"a": 10, "b": 25, "c": 15}
max_key = max(d, key=d.get)
print("Key with maximum value:", max_key)
```

5.
```
d = {"a": 12, "b": 22, "c": 43}
for key, value in d.items():
    print(key, ":", value)
```